EXPERIMENT NO. 3

Student Management using JSP, JDBC and MySQL

Simplified 8-file version | CRUD | No HTML output tables

What the manual asks for

Develop a JSP application that integrates JDBC to perform Create, Read, Update and Delete operations on a **students** database table, using roll number, name, email and age.

Files to remember

- **Student.java** - keeps one student record.
- **DBUtil.java** - opens the MySQL connection.
- **StudentDAO.java** - contains INSERT, SELECT, UPDATE and DELETE SQL.
- **index.jsp** - home navigation.
- **add-student.jsp** - Create.
- **list-student.jsp** - Read all records.
- **update-student.jsp** - loads one existing record and updates it.
- **delete-student.jsp** - deletes by roll number.

Study order

1. Memorize DBUtil connection details. 2. Learn the four DAO methods. 3. Remember each JSP calls the corresponding DAO method. 4. Practise the SQL setup and test Create, Read, Update, Delete in that order.

Database setup

```
CREATE DATABASE IF NOT EXISTS student;
CREATE USER IF NOT EXISTS 'student'@'localhost' IDENTIFIED BY 'student';
GRANT ALL PRIVILEGES ON student.* TO 'student'@'localhost';
FLUSH PRIVILEGES;
USE student;
CREATE TABLE IF NOT EXISTS students (
    roll_no INT PRIMARY KEY,
    name VARCHAR(150),
    email VARCHAR(150),
    age INT
);
```

The credentials must match DBUtil.java. Run the CREATE USER/GRANT statements as a MySQL administrator if the student user is not already configured.

Eclipse / Tomcat setup

- Create a Dynamic Web Project named **StudentApp**.
- Copy the Java files under **src/com/student/...** into the project source folder.
- Copy the JSP files into **WebContent** (or **src/main/webapp** for a newer layout).
- Add MySQL Connector/J 8.x or later to **WebContent/WEB-INF/lib**.
- Configure a Tomcat runtime and start MySQL.
- Run **index.jsp** on the server. Example URL: `http://localhost:8080/StudentApp/index.jsp`.

Expected results (sample)

- **Create:** Student added successfully!
- **Read:** list shows Roll No, Name, Email, Age, plus Update and Delete links for every record.
- **Update:** Student updated successfully! (or a not-found message if that roll number does not exist).
- **Delete:** Student deleted successfully! (or a not-found message if that roll number does not exist).

Student.java

Java source: model | Path: src/com/student/model/Student.java

```
package com.student.model;

public class Student {
    private int rollNo;
    private String name;
    private String email;
    private int age;

    public Student(int rollNo, String name, String email, int age) {
        this.rollNo = rollNo;
        this.name = name;
        this.email = email;
        this.age = age;
    }

    public int getRollNo() { return rollNo; }
    public String getName() { return name; }
    public String getEmail() { return email; }
    public int getAge() { return age; }
}
```

DBUtil.java

Java source: database connection | Path: src/com/student/util/DBUtil.java

```
package com.student.util;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class DBUtil {
    private static final String URL =
        "jdbc:mysql://localhost:3306/student?serverTimezone=UTC";
    private static final String USER = "student";
    private static final String PASSWORD = "student";

    public static Connection getConnection() throws SQLException {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            throw new SQLException("MySQL JDBC driver not found.", e);
        }

        return DriverManager.getConnection(URL, USER, PASSWORD);
    }
}
```

StudentDAO.java

Java source: CRUD SQL | Path: src/com/student/dao/StudentDAO.java

```
package com.student.dao;

import com.student.model.Student;
import com.student.util.DBUtil;
import java.sql.*;
import java.util.ArrayList;
import java.util.List;

public class StudentDAO {

    public void addStudent(Student s) throws SQLException {
        String sql = "INSERT INTO students VALUES(?, ?, ?, ?)";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, s.getRollNo());
            ps.setString(2, s.getName());
            ps.setString(3, s.getEmail());
            ps.setInt(4, s.getAge());
            ps.executeUpdate();
        }
    }

    public List<Student> getAllStudents() throws SQLException {
        List<Student> students = new ArrayList<>();

        try (Connection con = DBUtil.getConnection();
             Statement st = con.createStatement();
             ResultSet rs = st.executeQuery(
                 "SELECT * FROM students ORDER BY roll_no")) {

            while (rs.next()) {
                students.add(new Student(
                    rs.getInt("roll_no"),
                    rs.getString("name"),
                    rs.getString("email"),
                    rs.getInt("age")
                ));
            }
        }

        return students;
```

StudentDAO.java (continued)

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```
    }

    public Student getStudentByRollNo(int rollNo) throws SQLException {
        String sql = "SELECT * FROM students WHERE roll_no = ?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, rollNo);

            try (ResultSet rs = ps.executeQuery()) {
                if (rs.next()) {
                    return new Student(
                        rs.getInt("roll_no"),
                        rs.getString("name"),
                        rs.getString("email"),
                        rs.getInt("age")
                    );
                }
            }
        }

        return null;
    }

    public boolean updateStudent(Student s) throws SQLException {
        String sql =
            "UPDATE students SET name=?, email=?, age=? WHERE roll_no=?";

        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, s.getName());
            ps.setString(2, s.getEmail());
            ps.setInt(3, s.getAge());
            ps.setInt(4, s.getRollNo());

            return ps.executeUpdate() > 0;
        }
    }

    public boolean deleteStudent(int rollNo) throws SQLException {
        String sql = "DELETE FROM students WHERE roll_no = ?";
```

StudentDAO.java (continued)

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```
        try (Connection con = DBUtil.getConnection();
             PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, rollNo);
            return ps.executeUpdate() > 0;
        }
    }
}
```

index.jsp

JSP: home page | Path: WebContent/index.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<body>
    <h2>Student Management</h2>

    <a href="add-student.jsp">Add Student</a><br><br>
    <a href="list-student.jsp">View Students</a><br><br>
    <a href="delete-student.jsp">Delete Student by Roll No</a>
</body>
</html>
```

add-student.jsp

JSP: Create | Path: WebContent/add-student.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student" %>
<html>
<body>
    <h2>Add Student</h2>

    <form action="add-student.jsp" method="post">
        Roll No: <input type="number" name="rollNo" required><br><br>
        Name: <input type="text" name="name" required><br><br>
        Email: <input type="email" name="email" required><br><br>
        Age: <input type="number" name="age" required><br><br>
        <input type="submit" value="Add Student">
    </form>

    <%
        if ("POST".equalsIgnoreCase(request.getMethod())) {
            try {
                int rollNo =
                    Integer.parseInt(request.getParameter("rollNo"));
                String name = request.getParameter("name");
                String email = request.getParameter("email");
                int age = Integer.parseInt(request.getParameter("age"));

                Student s = new Student(rollNo, name, email, age);
                new StudentDAO().addStudent(s);

                out.println("<p>Student added successfully!</p>");
            } catch (Exception e) {
                out.println("<p>Could not add student: "
                    + e.getMessage() + "</p>");
            }
        }
    %>

    <br><a href="list-student.jsp">View Students</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

list-student.jsp

JSP: Read all records | Path: WebContent/list-student.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student,java.util.List" %>
<html>
<body>
    <h2>Student List</h2>

    <%
        try {
            List<Student> students =
                new StudentDAO().getAllStudents();

            if (students.isEmpty()) {
                out.println("<p>No students found.</p>");
            }

            for (Student s : students) {
    %>
                Roll No: <%= s.getRollNo() %><br>
                Name: <%= s.getName() %><br>
                Email: <%= s.getEmail() %><br>
                Age: <%= s.getAge() %><br>

                <a href="update-student.jsp?rollNo=<%= s.getRollNo() %>">Update</a> |
                <a href="delete-student.jsp?rollNo=<%= s.getRollNo() %>">Delete</a>

                <hr>
    <%
            }
        } catch (Exception e) {
            out.println("<p>Could not load students: "
                + e.getMessage() + "</p>");
        }
    %>

    <a href="add-student.jsp">Add Student</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

update-student.jsp

JSP: Update existing student | Path: WebContent/update-student.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO,com.student.model.Student" %>
<html>
<body>
    <h2>Update Student</h2>

    <%
        StudentDAO dao = new StudentDAO();
        Student student = null;
        String message = "";
        boolean post = "POST".equalsIgnoreCase(request.getMethod());

        try {
            if (post) {
                int rollNo =
                    Integer.parseInt(request.getParameter("rollNo"));
                String name = request.getParameter("name");
                String email = request.getParameter("email");
                int age = Integer.parseInt(request.getParameter("age"));

                Student s = new Student(rollNo, name, email, age);

                if (dao.updateStudent(s))
                    message = "Student updated successfully!";
                else
                    message = "No student found with that roll number.";
            } else {
                String id = request.getParameter("rollNo");

                if (id != null && !id.trim().isEmpty())
                    student = dao.getStudentByRollNo(
                        Integer.parseInt(id));
            }
        } catch (Exception e) {
            message = "Could not update student: " + e.getMessage();
        }

        if (!message.isEmpty()) {
            out.println("<p>" + message + "</p>");
        }

        if (!post) {
            if (student != null) {
```

update-student.jsp (continued)

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```
    %>
                <form action="update-student.jsp" method="post">
                    Roll No: <%= student.getRollNo() %><br><br>

                    <input type="hidden" name="rollNo"
                           value="<%= student.getRollNo() %>">

                    Name: <input type="text" name="name"
                        value="<%= student.getName() %>" required><br><br>

                    Email: <input type="email" name="email"
                        value="<%= student.getEmail() %>" required><br><br>

                    Age: <input type="number" name="age"
                        value="<%= student.getAge() %>" required><br><br>

                    <input type="submit" value="Update Student">
                </form>
    <%
            } else {
                out.println("<p>Student not found. Open the student list and select Update.</p>");
            }
        }
    %>

    <br><a href="list-student.jsp">Back to Student List</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```

delete-student.jsp

JSP: Delete by roll number | Path: WebContent/delete-student.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<%@ page import="com.student.dao.StudentDAO" %>
<html>
<body>
    <h2>Delete Student</h2>

    <%
        String rollNoValue = request.getParameter("rollNo");
        String message = "";

        if ("POST".equalsIgnoreCase(request.getMethod())) {
            try {
                int rollNo = Integer.parseInt(rollNoValue);

                if (new StudentDAO().deleteStudent(rollNo))
                    message = "Student deleted successfully!";
                else
                    message = "No student found with that roll number.";
            } catch (Exception e) {
                message = "Could not delete student: " + e.getMessage();
            }
        }

        if (!message.isEmpty()) {
            out.println("<p>" + message + "</p>");
        }

        if (rollNoValue == null)
            rollNoValue = "";
    %>

    <form action="delete-student.jsp" method="post">
        Enter Roll No:
        <input type="number" name="rollNo"
               value="<%= rollNoValue %>" required><br><br>
        <input type="submit" value="Delete Student">
    </form>

    <br><a href="list-student.jsp">Back to Student List</a><br><br>
    <a href="index.jsp">Back to Home</a>
</body>
</html>
```